MineXR: Virtual Mines for Safer Mining Education

M. Vallbé, M. Cardu, A. Gogolewska, G. Arbesser, J. Krysiak, M. Holly, B. Barbarzak, O. Pavlova, J. Pirker and J.J. Puente Sandoval

Problem Statement

1) **Safety Risks:**

- Extreme and hazardous environments.
- Rock falls, explosions, equipment accidents, exposure to harmful substances.
- Lack of experience in emergencies or in heavy machinery.

2) Limitating Conditions:

- Limited access to sites (remoteness, logistics, regulations, costs).
- Limited number of team partners.
- Environmental impacts of training.

Three killed after tunnel collapses in Spanish potash mine

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By **Euronews** with **AP**

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An official from the Spanish operator ICL Iberia confirmed that the three men were Spanish nationals and experienced geologists and that two of them were postgraduate students.

Proposed Solution

Enhance mining education safety practices and practical training through immersive technologies (VR, AR, MR).

XR improves engineering skillsets:

1) Practical experience
- Safety training (drills, operations, design, trial+error, …)
- 3D twins of mining environments (remote, expensive, unsafe)

2) Understanding of complex three-dimensional processes
- Improved engagement and retention (gamification, uncovering invisible phenomena)
- Widened learning styles

3) Knowledge of human-mine relationship
- Hands-on, interactive learning experiences that bridge the gap between theory and practice.
- Immediate application and feedback

4) Collaboration Skills
- Remote Collaboration and Learning

MineXR: Building Open XR Mines for Safe Enhanced Mining Education

Unite! Seed Fund for Teaching and Learning

2024 – 2025

Universitat Politècnica de Catalunya • Politechnika Wrocławska • Politecnico di Torino • Technische Universität Graz • Khmelnytskyi National University

Overview

«MineXR:

- transforms **mining engineering education** developing immersive **XR technology**,
- engaging students and teachers in innovative **safety training**, and
- fostering **collaboration** in a risk-free virtual environment for future engineers **across European universities.**»

Partnership

TU Graz

Wrocław University of Science and Technology

Politecnico di Torino

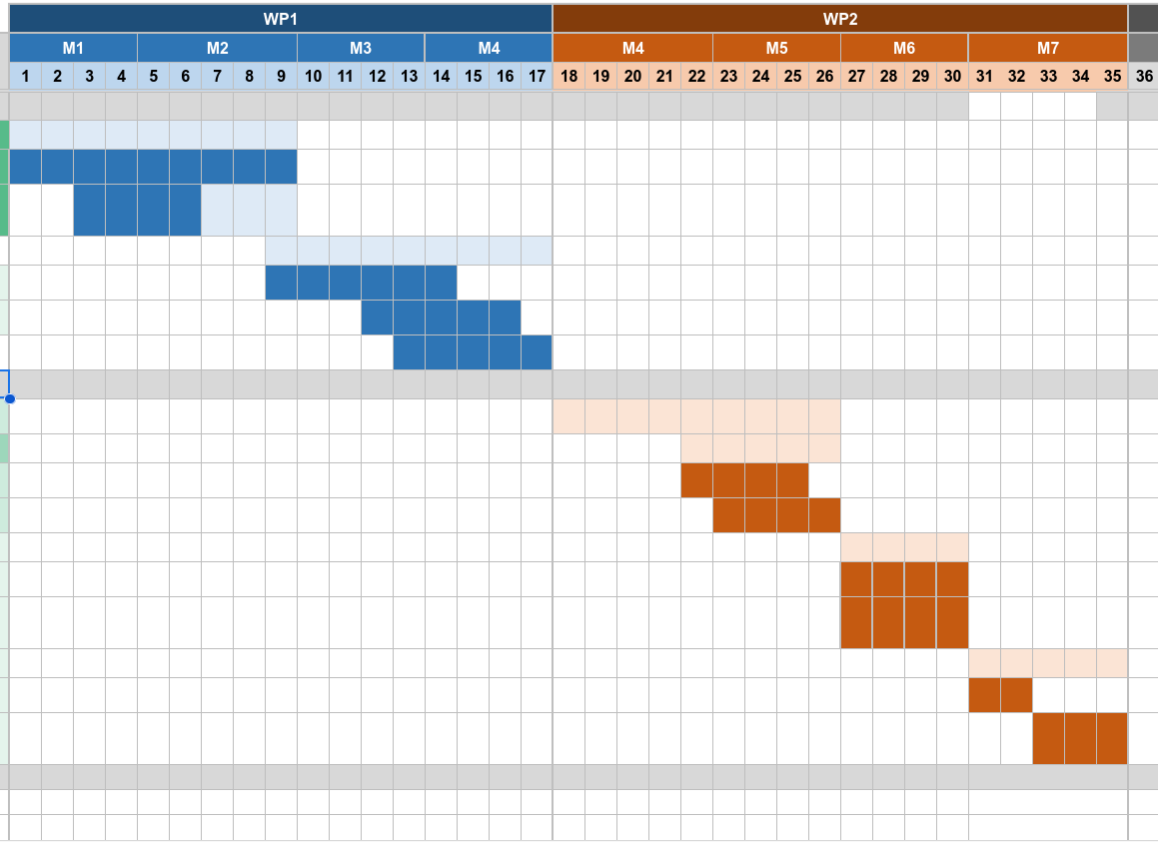
KHMELNYTSKYI NATIONAL UNIVERSITY

UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

MineXR GANTT CHART

PROJECT TITLE	Building Open XR Mines for Safe Enhanced Mining Education	UNITE! SEED FUND	Teaching and Learning
PROJECT ACRONYM	MineXR	DATE	2025

TASK ID	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
WP 1	Project Planning & Design		2024-12-01	2025-03-31	121	
1.1	Needs Assessment					100%
1.1.1	Identify specific safety concepts/scenarios to addres in the XR environment.					100%
1.1.2	Input from mining industry experts, safety professionals, educators, and students to determine the most critical safety topics.					100%
1.2	Content Design					40%
1.2.1	Storyboard outlining the sequence of events and scenarios to include in XR environment.					50%
1.2.2	Define interactive elements: hazards, safety equipment, emergency procedures, decision-making scenarios.					50%
1.2.3	Detail project plan: timelines, roles and responsibilities, communication protocols.					40%
WP 2	XR Mine Development & Content Creation		2025-04-01	2025-07-31	122	
2.1	Geophysical characterization of mines to inform the modeling and simulation process.					5%
2.2	Modeling & simulation of XR mines					10%
2.2.1	Create 3D models of mine structures, equipment, machinery, and geological features.					5%
2.2.2	Use XR development tools to build the virtual environment, incorporating 3D models, textures, lighting, and animations.					5%
2.3	Interactivity & Immersion					
2.3.1	Implement interactive elements to engage users and simulate real-world interactions.					
2.3.2	Include features such as interactive controls, equipment operation, hazard recognition, emergency response simulations, decision-making scenarios.					
2.4	Creation of Safety Training Educational Content					
2.4.1	Develop safety training interactive modules aligned with partner's curriculum objectives and learning outcomes.					
2.4.2	Integrate these modules seamlessly into the XR environment, ensuring that students can access relevant safety information and training resources as they navigate the virtual mine.					
WP 3	Pilot Testing & Evaluation		2025-08-01	2025-09-30	61	
3.1	Testing & Feedback					0%
3.1.1	Conduct usability, effectiveness, engagement, and satisfaction testing with target students, educators, and industry					0%



After needs assessment, we set upon content design:

- Safety in blasting learning
  - Drilling holes
  - Placing explosives inside those holes
  - Detonating the explosives
  - Clearing away displaced matter
- VR of an open pit mine
  - Simpler approach
  - 360° Camera?

Piggy-back on Game Lab Graz VRChances platform:

Deliverables

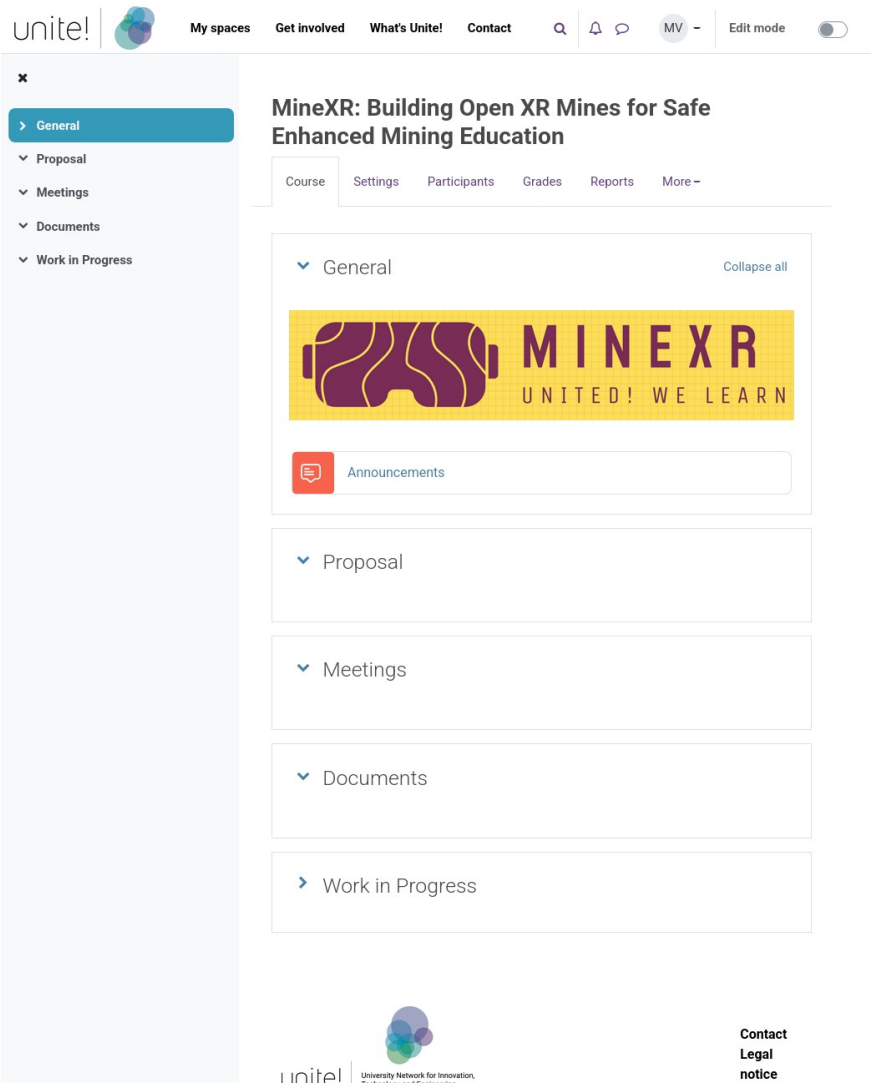
1. Needs Assessment report, useful for general engineering safety learning.
2. Minimal viable XR mine environment toolkit on Metacampus, including design documentation & technical specifications.
3. Pilot Testing report with feedback from students & instructors.
4. Evaluation report on the effectiveness of XR mine environments.
5. Final project report summarising achievements and recommendations.

Expected impact

1. Enhanced safety learning outcomes for mining engineering students.
2. Convergence in teaching methodologies across partner universities.
3. Follow-up proposals and other joint activities (COIL).
4. Contribution to the broader engineering education community through dissemination activities.

Sharing results with wider Unite! Community

- Metacampus: resource sharing, online forums and ongoing communication.
- Hybrid workshop to discuss results and plan further developments (COIL, XR, AI, safety for other engineering disciplines).
- Publications, presentations, and conference participation to showcase project outcomes and lessons learned.
- Long-term vision: scaling the project with a COIL across Unite! alliance.



Thank You

👆 get involved ❤️

unite! | University Network for Innovation, Technology and Engineering

A Unite!-Ed Future –

Reimagining Education for AI ready graduates

Teaching and Learning Unite! Conference 2025

Date: March 13, 9.00-16.00 CET
Venue: Zoom (hyperlinks in the headers)

Information about the conference: The 2nd Unite!-Ed Future Conference – Reimagining Education for AI ready graduates | Metacampus

Detailed Programme

9.00–9.10	**Introduction, Arnold Pears, zoom room**
9.10–10.00	**Keynote – *Beyond AI Hype and Toward Critical AI Education*, Matti Tedre and Henriikka Vartiainen zoom room**

10.00–10.15	**Information and short break, zoom room**

10.15–11.35	**Track 1** **Seed fund projects** **zoom room**	**Track 2** **Seed fund projects** **zoom room**	**Track 3** **Seed fund projects** **zoom room**
Session 1	***How do we teach math in Europe?*** (TU Graz) ***Reimagining Education for AI-Ready Graduates: Circular Economy Pathways and Energy Storage*** (ULisboa) ***Unite! Online Language Café*** (Aalto, Grenoble INP-UGA, TUDa, TU Graz, Wroclaw Tech)	***MineXR: Building Open XR Mines for Safe Enhanced Mining Education*** (PoliTO, TU Graz, UPC, Wroclaw Tech, KhNU) ***Collaborative Learning Environments for Research-based Teaching on Industry 4.0 (CLERT-I4.0)*** (Aalto, TUDa, ULisboa, UPC) ***UNEIT - Developing a Unite! Nuclear Education InitiaTive*** (Aalto, INP-Grenoble, PoliTO, ULisboa, UPC)	***Architriks: A Peer-Learning Platform for Tailored Digital Skill Development*** (KTH) ***Building OER Competence Across the Unite! Alliance: Insights from the "Unite! OER Courses" Project*** (Aalto, TU Graz)

11.40- 12.30	**Track 1** **Full paper/ work in progress** **zoom room**	**Track 2** **Work in progress** **zoom room**
Session 2	***Imagine There is a Human Learning*** (Aalto) ***The Quest for Urban Obdurate Resilience in Uncertain Milieus (QUORUM): An initial overview*** (Aalto, KTH, PoliTO, UPC, ULisboa)	***Defeating the free-rider problem in group work by mutual evaluation of the contribution to team effort*** (Aalto) ***The Metacampus as a Teaching Hub for the Unite! Alliance*** (TU Graz, UPC)

12.30-13.30	**Lunch**

13.35- 14.35	**Track 1** **Full paper/work in progress** **zoom room**	**Track 2** **Workshop** **zoom room**
Session 3	***Beyond Open Access: Open Educational Resources (OER) for Legal Clarity, Sustainability, and Digital Sovereignty in European University Alliances*** (TU Graz) ***Future Water: Online water education driven by climate change, a pandemic and the rise of generative AI*** (TU Graz, UArizona)	***Reflecting on Teaching in the Age of AI: A Tool for Critical Education*** (UPC)

14.40- 15.40	**Track 1** **Workshop** **zoom room**	**Track 2** **Workshop** **zoom room**
Session 4	***Promising Roles of AI in Education*** (Aalto)	***Reimagining Teacher Training: CollectiveGPT Workshop for the AI Era*** (TU Graz)

15.45-16.00	**Lightning talks and conference closing – Sven Bossuyt, zoom room**

A Unite!-Ed Future –

Reimagining Education for AI ready graduates

Teaching and Learning Unite! Conference 2025

Abstracts

Keynote speech

Beyond AI Hype and Toward Critical AI Education
H. Vartiainen and M. Tedre

As AI systems increasingly drive or mediate work, knowledge, and our daily lives, we face a crucial challenge: How do we prepare graduates who are not just AI users, but active citizens capable of understanding, challenging, and shaping the future of AI?

This talk focuses on the realities of AI and how education can respond to those. AI is not a "future technology"; it is already embedded in search engines, recommendation systems, fraud detection, and content creation. Yet, mainstream education still struggles to integrate AI literacy into curricula, often reacting to AI advancements with panic rather than preparation. We describe AI education that 1) helps learners build a scientific understanding of AI mechanisms, replacing folk theories with evidence-based knowledge; 2) supports them to develop critical data literacy, recognizing how AI-driven systems collect,
profile, and influence society; 3) engage them in AI design and inquiry, and to create, modify, and reflect on AI tools in personally meaningful ways; and 4) foster ethical and societal reflection, as well as helping them understand AI as part of society's power structures.

Session 1 – track 1

Moderator: Ernest Ampadu

How do we teach math in Europe?
G. Dorn

This Unite! Seed fund project focusses on the math education at universities in the first year and tries to improve onboarding by providing a MOOC (Fit for math).
The project partners also aim to share best practise and materials (OER) to improve the way we teach mathematics in Europe.

Problem statement:
Universities within the Unite! alliance phase a reduction of first-year students and high dropout years. This Seed Fund projects aims to provide a better onboarding on Unite! universities to reduce dropout rates especially in mathematics.

Background:
This Unite! Seed project is a follow up of the project "Freshman math skills and anxiety evaluation" which aimed to assess math anxiety and countermeasures in the first year.
The survey has been conducted within the Unite! network reaching approximately 10000 students of which 4000 started the survey, 3000 finished it, 2000 among them are first year students and 500 wrote individual comments.
A short overview of the results will be presented.

Method:
MOOC: Within the project we refactor an existing MOOC – massive open online course which is

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currently only available in German has a yearly subscription of 1000 students in Austria with a completion rate of 35% (evaluation figures of the current MOOC will be shown).
The seed fund project focuses to identify the common needed skills, enhance the existing MOOC, localize it (translation to 5-10 Unite languages) and make it accessible to all Unite! students via the Metacampus.
Workshops are organized at partner universities with the following topics:
- Meet the lecturers: the way we teach mathematics. A self-developed Canvas for math lecturers has been developed which will be shared.
- Ask the students: Meet local students to collect their experiences – a first workshop at 4th of December last year at TU Graz gave interesting results about the barriers, perception difficulties first-year students phase.
- Share local technologies: a format to collect best practices used in teaching at partnering universities

Results:
Expected results are a MOOC (onboarding in mathematics) with homogenized skills, translated content, and state of the art activation tools – such as h5p enhanced videos, manim animation and even implemented self-developed mathematical games.

Already existing result is the resume of the first two-days workshop from December 2024 in Graz.

Conclusion:
This seed funding activity aims to unite initiatives to improve onboarding at university especially in the field of mathematics.

Reimagining Education for AI-Ready Graduates: Circular Economy Pathways and Energy Storage
M. Alves, M. Taryba, F. Montemor

This work aims to transform education by equipping graduates with the knowledge and skills to thrive in AI-driven and sustainability-focused environments, particularly in electrochemical energy storage. By leveraging hybrid learning environments, incorporating student-centred pedagogies, and addressing sustainable development goals, the initiative aspires to redefine how students are prepared for the demands of an electrified and modern society. Central to this effort is developing an innovative e-course and a comprehensive compendium of knowledge, with a significant focus on teaching Circular Economy Pathways for the Energy Transition—a critical area for addressing global sustainability challenges.

A distinctive feature of the work is its emphasis on sustainable materials and design. The initiative will derive innovative, AI-supported research in biobased matrices for targeted applications. These materials use sustainable carbon sources to mitigate environmental pollution caused by metals and enhance CO2 assimilation, offering a dual benefit of sustainable development and environmental remediation.

We integrate case studies and challenge-based learning approaches to foster practical and analytical skills among learners. By aligning educational strategies with the pressing need for sustainable energy systems, we seek to enhance workforce preparation, ensuring a supply of professionals' adept at navigating circular economy principles that address the evolving needs of society, economy, and industry. Additionally, raising awareness among students and society about sustainability, circularity, and resource use in energy systems is critical for fostering an informed and proactive generation capable of addressing environmental challenges.

This work, carried out by a team from Instituto Superior Técnico, stems from the CESynergy Erasmus+ project, in collaboration with five other partners led by a Polish partner. By combining cutting-edge pedagogical approaches, interdisciplinary collaboration, and a focus on sustainability, the project sets a benchmark for international educational innovation and societal impact.

Unite! Online Language Café
R. Breit, B. Kjellgren, S. Belanger, I. Landsiedler, K. Raich, A. Mirowska and Y. Cho

Unite!'s multilingual and multicultural environment provides an excellent setting for informal language learning, which, according to language acquisition theories, can occur unconsciously and incidentally outside the classroom through interaction with native speakers or exposure to authentic language.

Our student-centred project, the virtual Unite! Online Language Café (U!OLC), aims to promote

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informal interactions that encourage students to use and learn all Unite! languages. The U!OLC is a virtual space where meetings are planned and organised by students for students.

In order to achieve our goals, we have developed didactic concepts, created training materials for student tutors, established an evaluation process and created a metacampus space where all relevant information and materials will be stored. Once completed, U!OLC will be a sustainable project that raises awareness of linguistic diversity and promotes autonomous learning.

Once operational, U!OLC will also provide a potential area for research on topics such as informal language acquisition, intercultural communication and the dynamics of peer-led online learning environments. Our model is a versatile and powerful addition to Unite!'s repertoire of educational offerings, highly adaptable and easily transferable to other contexts and audiences.

In the process, the project has provided valuable insights into the challenges of translating a traditionally face-to-face activity into an engaging online format. These lessons, particularly around participation, interaction and technology, will be shared during our presentation.

Session 1 – track 2

Moderator: Arnold Pears

MineXR: Building Open XR Mines for Safe Enhanced Mining Education

M. Vallbé, M. Cardu, A. Gogolewska, G. Arbesser, J. Krysiak, M. Holly, B. Barbarzak, O. Pavlova, J. Pirker and J.J. Puente Sandoval

To reduce injuries and fatalities among mining engineering students during practical training, we are developing an immersive eXtended Reality (XR) underground mine environment. This innovative tool will provide students pursuing bachelor's and master's degrees in mining engineering with a safe and realistic virtual space to practice essential skills and learn about safety procedures.

Our goal is to create a shared, open toolkit for mining engineering education across the Unite! network of universities. This will involve:

- Developing an XR mine environment: Simulating various mining operations, including exploration, extraction, and processing, with a strong emphasis on safety.

- Enhancing student learning: By providing hands-on experience in a safe virtual setting, improving engagement, and enhancing learning outcomes.

- Building an open educational resource: Creating a repository on Metacampus to share XR mining simulations, best practices, and resources with other Unite! colleagues and beyond.

- Facilitating international collaboration: Enabling students from different Unite! universities to train together in the same virtual environment through Collaborative Online International Learning (COIL) programs.

This interdisciplinary Seed Fund project brings together researchers from five universities with expertise in mining, computing, pedagogy, and remote sensing (Universitat Politècnica de Catalunya, Wrocław University of Science and Technology, Politecnico di Torino, Graz University of Technology, and Khmelnytskyi National University). We are currently in the early stages of developing a proof-of-concept prototype, and we believe this project will significantly advance safety training and prepare future mining engineers for the challenges of the industry.

Collaborative Learning Environments for Research-based Teaching on Industry 4.0 (CLERT-I4.0)

J. Minguella Canela, M. Leite, J. Metternich, V. Eloranta and J. Romeu Garbí

The CLERT-I4.0 initiative, funded by "Unite! Seed Fund" for Teaching and Learning, aims to create, deploy and use collaborative learning environments for research-based teaching in Industry 4.0 technologies between the Unite! Partner universities.

The project seeks to strengthen the collaboration between the Universitat Politècnica de Catalunya (UPC), the Instituto Superior Técnico-Universidade de Lisboa (IST-UL), the Technical University of Darmstadt (TUDa) and Aalto University in teaching and research in Industry 4.0. This collaboration will take the form of joint research assignments for students enrolled in different masters of the participating universities, that can combine the expertise of each university

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and research groups, including laboratory-intensive short international mobility stays. In this context, CLERT-I4.0 seeks to establish a framework for research-based collaborative teaching in Industry 4.0, with a special focus on 3D printing. To achieve this, international learning formats will be deployed, co-created by the partners, with a strong base in laboratories and assisted by online tools. In addition, short research stays of 2 months will be carried out in a host institution, with the possibility of virtual mobility when physical mobility is not viable.

In CLERT-I4.0, the Unite! Metacampus will be used as a central learning management platform for all members, including trainings, workshops and lectures. Most meetings will be held online to be more cost-efficient, with monthly meetings throughout the project. The project plan includes synergistic preparatory work, implementation of research assignments, and project management and dissemination for 11 months. Research assignments will be made in collaboration between the participating universities, with a strong base in laboratories and assisted by online tools. These assignments will be carried out partly at the original institution and partly at the host institution, with short mobility stays of 2 months.

Expected outcomes include interdisciplinary collaboration, a collaborative framework, research allocations and dissemination factors. CLERT-I4.0 also seeks to promote attractive thesis topics for potential new candidates and ensure the sustainability of joint research-based teaching. Also, it is interesting to mention that, although this project focuses on advanced manufacturing, the concept but could be applied to other areas such as sustainability, energy, and entrepreneurship. Moreover, the initiative can find sustainability using Erasmus+ Internship grants for the students.

In summary, CLERT-I4.0 initiative aims to create an international collaboration of special interest that can be the seed for a sustained and sustainable partnership between research and teaching actors with common interests and goals.

Session 1 – track 3

Moderator: Verena Schwägerl-Malchior

Architriks: A Peer-Learning Platform for Tailored Digital Skill Development

K. Brahma

In an era where Artificial Intelligence (AI) and digital tools are deeply embedded in academic and professional workflows, digital learning platforms play a critical role in developing skill-ready graduates. This abstract explores a personalized peer-learning model where university seniors teach relevant software skills tailored to specific academic disciplines, creating a more engaging and effective learning environment compared to generic online platforms.

The idea originated from an online Photoshop for Architects class I conducted for first and second year students at my undergraduate college in India. By focusing on the software's application in architecture-specific projects, the class addressed practical, contextual needs. The alumni-student connection fostered trust, enabling students to ask personal, course-related, and career-oriented questions. This human interaction and peer-to-peer motivation are often lacking in platforms like YouTube, which provide free but impersonal tutorials.

The success of this initiative has inspired a scalable concept – "Architriks", a website designed to connect skilled university seniors with juniors seeking affordable, tailored software training. Architriks builds on the principles of student-centered learning, peer interaction, and accessibility. By focusing on academic context and individual needs, it supports AI-ready graduates through the development of digital tools, specialized workflows, and sustainable learning practices.

The platform aims to address two key challenges: first, the lack of contextualized digital learning resources for students in specific fields, and second, the absence of motivation and mentorship in traditional online learning spaces. Architriks bridges this gap by encouraging collaborative knowledge exchange and creating opportunities for seniors to share expertise while earning.

This platform will target students in their first and second year of education and connect them with

final year students in their field from Swedish universities(or global) and enable them to learn more about the field directly from their alumni. It will consist of a volunteer system where final year students volunteer to sign up for the program and create an online profile and the students in their first two years can choose their mentor to get guidance from. A lot of work will go into creating the wed design for this platform and gathering data of interested volunteers.

This model aligns with the themes of interaction in digital learning environments, student-centered teaching, and sustainable development of educational practices. By fostering peer-to-peer mentorship and targeted skill development, Architriks can empower students to become more adaptable and AI-ready in their respective disciplines.

Building OER Competence Across the Unite! Alliance: Insights from the "Unite! OER Courses" Project
S. Schön, M. Ebner and R. Vicente-Sáez

The "Unite! OER Courses" seed fund project successfully advanced Open Educational Resources (OER) competence among students and lecturers across the Unite! alliance. A major milestone was the development of a Massive Open Online Course (MOOC) on OER, licensed under CC BY 4.0 International and translated into 11 languages, including Turkish and Indonesian, with the help of AI avatars and tools. Since its launch on May 6, 2024, the MOOC has attracted over 1,200 participants. Additionally, an OER course in English was introduced on the Unite! federated learning management system Metacampus.

To further enhance capacity, a dedicated OER training for lecturers was conducted, which resulted in the publication of a collection of tools and methods for future OER training initiatives. In this contribution, we build upon previous work, particularly insights from our project report (Unite, 2024) and various presentations on the topic, including preliminary findings on the impact at partner universities (Schön et al., n.d., submitted for JODDE).

Methodologically, we addressed different research questions primarily through a case study, allowing us to analyse the implementation, reception, and institutional effects of the OER initiatives within the alliance. The collected data provides valuable insights for the establishment of a future Unite! Open Science Academy.

This presentation will share the project's achievements, methodologies, and outcomes while addressing challenges and opportunities for promoting OER practices in international higher education settings.

Session 2 – track 1

Moderator: Anna Wozna

Imagine There is a Human Learning
T. Kauppinen

Today we are already living with AI. But do we know what does it think about education? Therefore, let's have a conversation with AI.

Me: AI, please answer and ask a follow-up question: "what is the most important design element that needs to be introduced to make education future-proof?"
AI: The most important design element is adaptability, ensuring education systems can evolve with technological advancements and societal needs. How can we balance teaching foundational skills with preparing students for emerging, unpredictable future challenges?
Me: I argue that learning foundational skills such as reading, writing, mathematics, spatial thinking and empathy together with media literacy, challenge-solving, programming, visualization, language skills—or more abstractly the famous four Cs, i.e creativity, communication, critical thinking and collaboration all greatly prepare students to cope with the future challenges. But the answer given by AI—adaptability—is a very generic answer, almost a non-answer. When in history we should not have ensured that education systems have adaptability?
Undeniably, we must adapt again, and perhaps with greater urgency. Therefore I will contribute via the following three perspectives, all which build on humans learning:

1. Discussion on how my own teaching of information visualization at two universities, Aalto University and Universität Münster is supporting learners to learn not just skills but also mindsets to cope with the grand challenges of the world.

2. Use of immersive realities to enable engaging learning experiences for learners, where being a human is an asset and thus not replacable by AI.
3. Sharing ways of collaborating at international level to understand and contribute to the development of models, approaches and tools for about AI in education.

Firstly, the course Explorative Information Visualization is an adventurous learning experience as a joint journey into visual storytelling. Students start their exploration by identifying a potential topic related to United Nations' Sustainable Development Goals. This will be theme they plan to create a visualization about. Through engaging sessions about available approaches, reflections, individual work and group discussions they are ready to present their visualizations at the demo days for an interested audience.

Secondly, studying, creating and making use of mixed realities, including meta-verse, telepresence, podcasts and other forms of augmented and virtual reality enable creating learning experiences that stick with the learner. In my teaching and in our projects students can meet, collaborate and learn in 360°learning environments, walk in a park while listening to a podcast, reflect their learning by recording another and will peer review each other's works.

Thirdly, I believe that educators will similarly need to engage with diverse ways to learn, discuss and share their visions about the future of learning. For me this means having podcast discussions, editing and authoring both academic works and speculative fiction, and participating in panels, forums and initiatives to jointly understand how to create a truly fruitful educational environment in this AI era.

Summa summarum: it is crucial to imagine there is a human learning when studying and designing learning experiences.

The Quest for Urban Obdurate Resilience in Uncertain Milieus (QUORUM): An initial overview
C. Mady, E. Comino, A. Clua Uceda, A. Pears, S. Barreiros Proença, A. Botero and P. Pinho

This presentation introduces the Quest for Urban Obdurate Resilience in Uncertain Milieus (QUORUM) project. The project develops a case study module for UNITE! Partner universities using a learner-centred approach. The focus is on activities that contribute to the development of interdisciplinary competencies for future professionals facing complex urban issues. In particular, the module addresses water-related complex urban challenges, focusing on urban rivers as ecosystem service providers.

The increasing frequency of natural disasters, and the importance of considering nature-based solutions – especially in relation to water and climate change – emphasise the need to consolidate interdisciplinary collaboration in achieving transformative sustainable solutions. The project received funding from the UNITE! call with the focus area "Sustainable Cities and Water" and started in January 2025. During the project period, the participating experts and students will explore a pedagogical framework that develops students' skillsets for interdisciplinary, multicultural learning experiences. The goal is to enable them to effectively address water-related challenges amidst climate change within uncertain milieus.

The project team includes experts from five Unite! partner universities: Aalto University, University of Lisboa, Politecnico di Torino, Universitat Politècnica de Catalunya, and Royal Institute of Technology (KTH). In the co-design of this graduate level module, the team has set three objectives; first, to explore interdisciplinary, multicultural, and collaborative pedagogical methodologies used within courses addressing urban rivers as ecosystem services at the team's universities; second, to develop specific ideas for co-designing a graduate-level student module (3 ECTS) on urban rivers as ecosystem services; and third, and to communicate project outcomes, and joint approaches to engaging students in interdisciplinary, multicultural, and collaborative pedagogical methodologies. The outcomes are also intended to be offered within UNITE! and the wider educational community. Three work packages address the project objectives. The first is about collecting pedagogical approaches on interdisciplinary, multicultural learning

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environments dealing with rivers in the teams' universities; the second involves co-designing the urban river module; and the third relates to sharing the outcomes with UNITE! Community.

In this paper, the team presents its initial work related to the first objective and first work package and outlines the other two objectives. Through online meetings, the team has exchanged knowledge and experience on pedagogical frameworks and conducted a literature review on interdisciplinary and multicultural pedagogical approaches. Preliminary findings indicate the need to facilitate forming a common lexicon among participants, maintaining motivation across learners with diverse backgrounds, designing activities to facilitate equal participation, emphasising the significance of continuous peer-learning and individual reflection, and integrating social activities to create a sense of community among learners. Drawing on work from the research area of Project and Problem Based Learning, interactive scenario-based educational methods are integrated with pedagogical concepts from the Open-Ended Group Project (OEGP) literature. Future work includes co-design activities with student groups, and development of learning materials and scenarios.

Session 2 – track 2

Moderator: Isabel Ferreirim

Defeating the free-rider problem in group work by mutual evaluation of the contribution to team effort

S. Bossuyt and E. Tai Bossuyt

Many students dislike group work in courses because of the free-rider problem: teammates who did not contribute much to the effort stand to benefit from the efforts of those who did contribute. We have resolved this issue by applying a correction to the individual grades earned for the group work, based on mutual evaluation of the amount, the quality, and the timeliness of each member's contributions to the group work, as well as their communication with the team about it. In addition to enhancing the perceived fairness of the grades earned for group work, it motivates students to work together better as a team, and the evaluation process guides reflection on how to work together better as a team. We will describe our experiences implementing this in a project course and share some implementation details that proved important to ensure the fairness of the correction procedure.

The Metacampus as a Teaching Hub for the Unite! Alliance

K. Gasplmayr, S. Schön, J. Alcober, M. Ebner and X. Molins

The Unite! alliance has developed the Metacampus, a federated LMS based on Moodle, to facilitate seamless collaboration and resource sharing among partner universities. This presentation showcases how the Metacampus supports teaching and learning across the alliance by addressing diverse needs and enabling innovative approaches. Key use cases where the Metacampus proves particularly beneficial are outlined, along with resources created to guide users, including detailed documentation, courses, and instructional videos (produced with AI avatars). Activities such as targeted training sessions and the streamlined submission of requests and special requirements via an integrated request form are highlighted. LTI pilot activities, aimed at integrating local LMS systems with the Metacampus, are presented, along with insights from regular E-Learning Support Group meetings. Success stories demonstrate the platform's impact and potential for enhancing collaboration within Unite!. In conclusion, challenges extending beyond technical solutions are addressed, such as the lack of incentives to adopt Unite!-specific courses or to fully utilize the Metacampus. These insights foster dialogue on how to further support teaching and learning within the Unite! alliance.

Session 3 – track 1

Moderator: Ernest Ampadu

Beyond Open Access: Open Educational Resources (OER) for Legal Clarity, Sustainability, and Digital Sovereignty in European University Alliances

M. Ebner and S. Schön

Open Educational Resources (OER) are widely recognized for improving access to education and enabling the sharing of knowledge. However, in the context of European university alliances, such as Unite!, OER offer additional, often underappreciated benefits that are crucial for cross-border collaboration and sustainable development in higher education. This position

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paper explores three key aspects of OER that are particularly relevant for European alliances. First, OER enable the legally secure use of educational resources across national borders, addressing uncertainties about copyright laws, particularly for translations and adaptations. This ensures compliance with different legal frameworks while fostering collaboration. Second, OER support sustainability by ensuring that investments in educational materials are not limited by restrictive usage rights. This is particularly critical in alliances where shared resources play a central role in fostering long-term cooperation and aligning with sustainability goals, a priority for Unite!. Finally, OER contribute to digital sovereignty by empowering institutions and educators to create, adapt, and share resources without reliance on proprietary platforms or licenses. This aligns with the broader strategic objectives of European alliances to promote autonomy and resilience in their digital ecosystems. By highlighting these often-overlooked benefits of OER, this research aims to broaden the perspective on their strategic importance in fostering collaboration, sustainability, and sovereignty within European university alliances.

Future Water: Online water education driven by climate change, a pandemic and the rise of generative AI
M. Pointl, N. Lorber and K. Lansey

Challenges posed by climate change, urbanization, overuse of resources, and aging infrastructure drive a global demand for civil and environmental engineering (CEE) graduates equipped to develop innovative water management approaches. Traditional engineering curricula, however, rarely provide the essential skills needed to develop sustainable, transdisciplinary solutions. 'Future Water', an international web seminar, currently in its second iteration, is an open education initiative, designed to support CEE students in advancing crucial communication and collaboration skills in an interdisciplinary setting.

In Future Water, students develop and present holistic solutions for predefined 'water management challenges' in self-organized, international teams, supported by lectures and input from faculty advisors. 'Challenges' define a number of characteristics for a town or region, including water availability, expected climate change, relative location to water sources, population, and community attitudes on growth, environmental stewardship etc. The basic concept is the same for both seminar iterations; the status quo of academic education, however, has evolved considerably since the first seminar. At Future Water's inception, virtual collaboration was a novel concept, placing its implications for collaborative learning at the focus of the seminar's evaluation. The first seminar concluded just before the COVID-19 pandemic made remote learning the norm for engineering programs. The recent, yet widespread adoption of generative AI by students and educators brought another disruption in CEE education, prompting further adaptations to Future Water's didactic concept and evaluation methodology.

This work examines how changes in the educational landscape and the rise of AI tools have influenced the conceptualization of both Future Water seminars - beyond the logistical challenges of running a virtual class across different academic systems and time zones. To this end, findings from the first seminar [1] are analyzed alongside preliminary data from the second, including lecture recordings, transcripts, student's work products, as well as surveys, time sheets, and AI prompts. This integrated analysis is designed to identify patterns, differences and gaps in student teams' water management solutions, which we aim to correlate to academic background, seniority, group dynamics, and cultural communication characteristics. Additionally, we intend to quantify AI usage in tackling water management challenges via the submitted prompts. Findings on collaboration, communication, and group dynamics will support the continuous adaptation of Future Water's concept as the requirements for water management education and CEE practice evolve.

Session 3 – track 2

Moderator: Anna Gogolewska

Reflecting on Teaching in the Age of AI: A Tool for Critical Education
D. Rocio Ramirez Rodriguez

In a world where Artificial Intelligence (AI) redefines the way we access and process information, it is imperative to rethink teaching in academia, and in this case to graduate students. Teaching how to handle AI properly is essential, but even more critical is learning how to interpret

it, evaluate it, and foster critical thinking in the face of the vast amount of immediate information it offers.

One relevant strategy that can be implemented in the classroom is using the "Reflective Writing" system proposed by Moon (2004). This approach encourages deep and reflective analysis through four progressive levels:

- Level 1: Descriptive Writing. Descriptive writing that focuses on recording facts.
- Level 2: Descriptive Reflection. Descriptive Reflection that adds basic analysis.
- Level 3: Dialogic Reflection. Dialogic Reflection that includes different perspectives and questioning.
- Level 4: Critical Reflection. Critical Reflection that delves into more complex implications and meanings.

To facilitate this process, teachers can use a critical reading checklist, with questions that guide students through each level of reflection. For example:
- What is the main point of the theme or innovation?
- Who makes it?
- What does it suggest?
- For whom is this topic important?

This approach allows not only to rescue ideas generated by AI, but also to encourage collaborative and reflective thinking by students, adapting to both face-to-face and hybrid environments.

In addition, the method can be scaled to address critical issues such as sustainable development. Inviting students to question and debate from diverse perspectives encourages the creation of scalable, multidisciplinary solutions.

This strategy develops skills such as leadership, teamwork, and critical thinking and promotes active participation in the classroom, turning AI into a tool that enhances learning rather than just a source of information.

Session 4– track 1

Moderator: Verena Schwägerl-Melchior

Promising Roles of AI in Education
T. Kauppinen

Given the emerging curiosity among educators and students about AI, it is now timely to workshop on promising roles of AI in education. Some of the roles are already in use, such as AI-powered assistants, personalized learning materials, feedback providers and quizz generators. In this workshop these and other maintstream roles are first presented. Inspired by examples participants will take a deep dive into workshopping and ideating what other roles AI can and should be given to improve learning and teaching experiences, help in designing programmes or even for running a higher education institution.

Schedule of the Workshop on Promising Roles of AI in Education

0-10 Introduction round of participants via an icebreaker
10-20 Brief presentation about the emerging roles of AI and tools
20-40 Group working session
40-55 Round of results sharing
55-60 Wrap-up

Session 4 – track 2

Moderator: Dominik Ruffeis

Reimagining Teacher Training: CollectiveGPT Workshop for the AI Era
B. Brünner

As generative AI (genAI) becomes a transformative force in education, it is crucial to improve educators' AI literacy in order to prepare students for a future shaped by artificial intelligence. This workshop, presented at the 2nd Unite!-Ed Future Conference, introduces strategies for understanding AI concepts and creating interactive, synchronous online learning experiences tailored to the needs of educators. By focusing on genAI-literacy, the session offers practical insights into understanding AI tools, with a focus on collaboration, and real-world teaching applications.
The workshop is guided by design-based research (DBR) methodology, which ensures iterative refinement and alignment with AI literacy

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frameworks and educational theory. Participants will work with a customized interactive platform, CollectiveGPT, that fosters real-time exploration of generative AI processes, including probabilistic modeling, context-dependence, bias and hallucinations. Through the collaborative activities, participants will gain a deeper understanding of how AI systems generate output and learn to use tools for effective integration into classroom practice.

Key concepts covered in the workshop include simplifying the fundamentals of AI by examining how generative AI predicts text using statistical probabilities and context, with examples rooted in educational content to make the concepts accessible. Participants will be introduced to prompt techniques, how role-based, style-specific, and emotion-driven prompts are used to improve AI outputs for practical applications in the classroom. The workshop also addresses data awareness and bias, highlighting how training data shapes AI behavior, revealing potential biases, and equipping participants with strategies for critically engaging with AI-generated content. In addition, the session addresses ethical and pedagogical challenges, such as prompt vulnerabilities and the importance of promoting transparency in AI systems. It emphasizes critical thinking to help educators effectively navigate the ethical dimensions of integrating AI into educational contexts.
The session follows a structured approach, progressing from basic AI concepts to advanced topics, ensuring accessibility for participants with varying levels of technical expertise. Pilot events in Austria showed the impact of the workshop: Most participants report a significant increase in their understanding of generative AI.
By addressing both the technical and pedagogical aspects of AI, this workshop equips educators with the tools to effectively integrate generative AI into their teaching practice. It supports the mission of the conference to reimagine education for AI-ready graduates and provides a structured framework for participants to tackle the challenges and opportunities of AI in education.

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